

Manual Derivation: Huggett Discrete-State Problem

ECONS 509 – Problem Set 1

Household problem. For current assets k and labor-efficiency state $s \in \{1, 2\}$,

$$v(k, s) = \max_{k' \geq \underline{k}} \left\{ u(c) + \beta \sum_{s'=1}^2 P_{ss'} v(k', s') \right\}, \quad c = (1 + r^0)k + w^0 \epsilon_s \bar{l} - k',$$

where $u(c) = c^{1-\sigma}/(1-\sigma)$ and, in the discrete problem, $k' \in \mathcal{K} = \{k_1, \dots, k_N\}$. The prices are fixed at

$$w^0 = (1 - \alpha)z^* \left(\frac{K^0}{L^*} \right)^\alpha, \quad r^0 = \alpha z^* \left(\frac{K^0}{L^*} \right)^{\alpha-1} - \delta,$$

with $L^* = \sum_s \pi_s^* \epsilon_s \bar{l}$.

Discrete-state vector form. Let $S = 2$ and $M = NS$. Store the value function as $\mathcal{V} \in \mathbb{R}^{N \times S}$, with column s equal to the values on the asset grid for shock s , and stack columns so that $m(n, s) = (s-1)N + n$. Thus $\mathcal{V}(\cdot) \in \mathbb{R}^M$. For current joint state $m(n, s)$ and candidate next-asset index j , define

$$c_{mj} = (1 + r^0)k_n + w^0 \epsilon_s \bar{l} - k_j, \quad \mathcal{R}_{mj} = \begin{cases} \frac{c_{mj}^{1-\sigma}}{1-\sigma}, & c_{mj} > 0, \\ -\infty, & c_{mj} \leq 0. \end{cases}$$

Hence $\mathcal{R} \in \mathbb{R}^{M \times N}$. Since $\mathcal{P}\mathcal{V}^T \in \mathbb{R}^{S \times N}$, the Kronecker product $(\mathcal{P}\mathcal{V}^T) \otimes \mathbf{1}_N$ is $M \times N$. The Bellman equation is

$$\mathcal{V}(\cdot) = \max_{j=1, \dots, N} [\mathcal{R} + \beta(\mathcal{P}\mathcal{V}^T) \otimes \mathbf{1}_N],$$

where the maximum is row by row. The policy $\mathcal{G} \in \{1, \dots, N\}^{N \times S}$ stores the maximizing asset index,

$$\mathcal{G}_{ns} = \arg \max_j \left\{ \mathcal{R}_{m(n,s),j} + \beta \sum_{s'} P_{ss'} \mathcal{V}_{js'} \right\}.$$

Given $\mathcal{G}$, define the policy transition matrix $\mathcal{Q} \in \mathbb{R}^{M \times M}$ by

$$\mathcal{Q}_{m(n,s),m(j,s')} = \begin{cases} P_{ss'}, & j = \mathcal{G}_{ns}, \\ 0, & \text{otherwise.} \end{cases}$$

If $\mathcal{R}^* \in \mathbb{R}^M$ collects flow utility at the chosen policy, fixed-policy evaluation satisfies

$$\mathcal{V}(\cdot) = \mathcal{R}^* + \beta \mathcal{Q} \mathcal{V}(\cdot), \quad (I - \beta \mathcal{Q}) \mathcal{V}(\cdot) = \mathcal{R}^*.$$

Continuous-choice Euler equation and borrowing constraint. For the continuous-choice version, write the constraint as $k' - \underline{k} \geq 0$ with multiplier $\lambda \geq 0$. Using the resource constraint to substitute for consumption, the Lagrangian is

$$\mathcal{L} = u((1 + r^0)k + w^0 \epsilon_s \bar{l} - k') + \beta \sum_{s'} P_{ss'} v(k', s') + \lambda(k' - \underline{k}).$$

The first-order condition for k' and the envelope condition are

$$-u'(c) + \beta \sum_{s'} P_{ss'} v_k(k', s') + \lambda = 0, \quad v_k(k, s) = (1 + r^0)u'(c).$$

Combining them gives the KKT Euler condition

$$u'(c) = \beta(1 + r^0) \sum_{s'} P_{ss'} u'(c'(s')) + \lambda,$$

with

$$\lambda \geq 0, \quad k' - \underline{k} \geq 0, \quad \lambda(k' - \underline{k}) = 0.$$

For CRRA utility, $u'(c) = c^{-\sigma}$. Therefore, whenever the borrowing constraint is slack ($k' > \underline{k}$), $\lambda = 0$ and

$$c^{-\sigma} = \beta(1 + r^0) \sum_{s'} P_{ss'} [c'(s')]^{-\sigma}.$$

Euler residual used in the accuracy check. Let the discrete policy be $g(k, s)$ and define policy consumption

$$c_g(k, s) = (1 + r^0)k + w^0 \epsilon_s \bar{l} - g(k, s).$$

At a slack state, put $k' = g(k, s)$ and, for each s' , let $k'' = g(k', s')$ so that $c_g(k', s')$ is next-period consumption. I use the consumption-equivalent relative Euler residual

$$e(k, s) = 1 - \frac{[\beta(1 + r^0) \sum_{s'} P_{ss'} c_g(k', s')^{-\sigma}]^{-1/\sigma}}{c_g(k, s)}.$$

The continuous-choice Euler equation implies $e(k, s) = 0$. In Question 2(h) I report $|e(k, s)|$ only where $g(k, s) > \underline{k}$; binding states obey the KKT inequality rather than the Euler equality.